Predicting Baseflow Response to Irrigation Pumping

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CS 5830

Introduction

Groundwater is a critical resource in the Republican River Basin, which spans parts of Colorado, Nebraska, and Kansas. Irrigation pumping from the underlying aquifer supports agriculture across the region, but heavy pumping can reduce baseflow. Declining baseflow threatens water availability for downstream users and ecosystems. This project uses multiple linear regression to analyze how irrigation pumping, precipitation, and evapotranspiration affect observed baseflow across 42 river segments in the basin.

Our analysis shows that pumping has a negative effect on baseflow, while precipitation increases it and evapotranspiration decreases it. Using these relationships, we derive climate-adaptive safe pumping thresholds that adjust based on weather conditions. These thresholds could help water managers set sustainable pumping limits that protect streamflow while still allowing irrigation. We used data from the Republican River Compact Administration (RRCA) groundwater model and applied ordinary least squares (OLS) regression with interaction terms to capture how climate moderates the effect of pumping.

Presentation Slides: https://docs.google.com/presentation

GitHub Repo: https://github.com/Grumblehead84/CS5830_Project6

Dataset

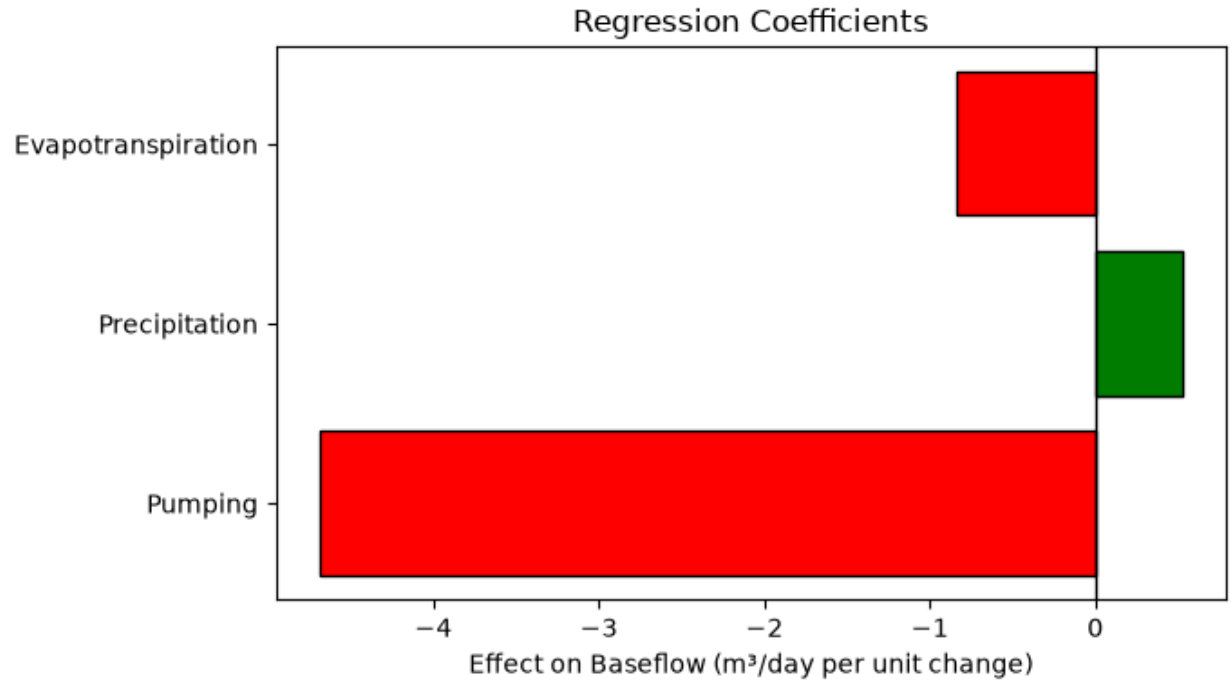
The dataset comes from the RRCA groundwater model and contains 15,591 records across 42 river segments. Each record includes the date, segment ID, geographic coordinates (x, y), and four key measurements: evapotranspiration, precipitation, irrigation pumping, and observed baseflow. The data spans multiple years, giving us enough observations to capture seasonal and spatial variation. This dataset is well-suited for regression analysis because it provides continuous numeric variables with clear physical relationships. For example, pumping depletes groundwater that feeds streams, precipitation recharges it, and evapotranspiration draws moisture from the soil.

Analysis Technique

We used multiple linear regression (OLS) because our goal is to understand how several continuous variables jointly affect baseflow. The model includes irrigation pumping, precipitation, and evapotranspiration as main predictors, plus two interaction terms (Pumping × Precipitation and Pumping × ET) to capture how climate conditions change the impact of pumping. We also included dummy variables for each of the 42 river segments to account for location-specific differences in geology and hydrology. Regression is a natural fit here because the relationships are approximately linear, the variables are continuous, and the model coefficients directly tell us the direction and magnitude of each factor's effect on baseflow.

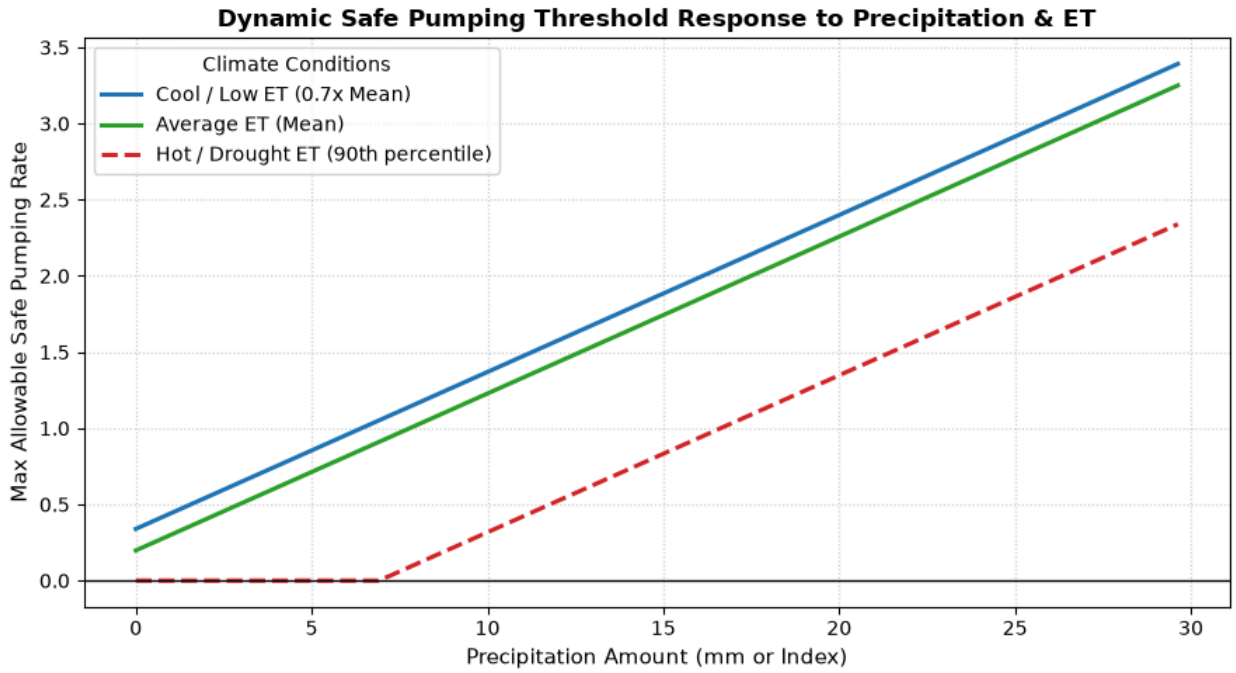
Results

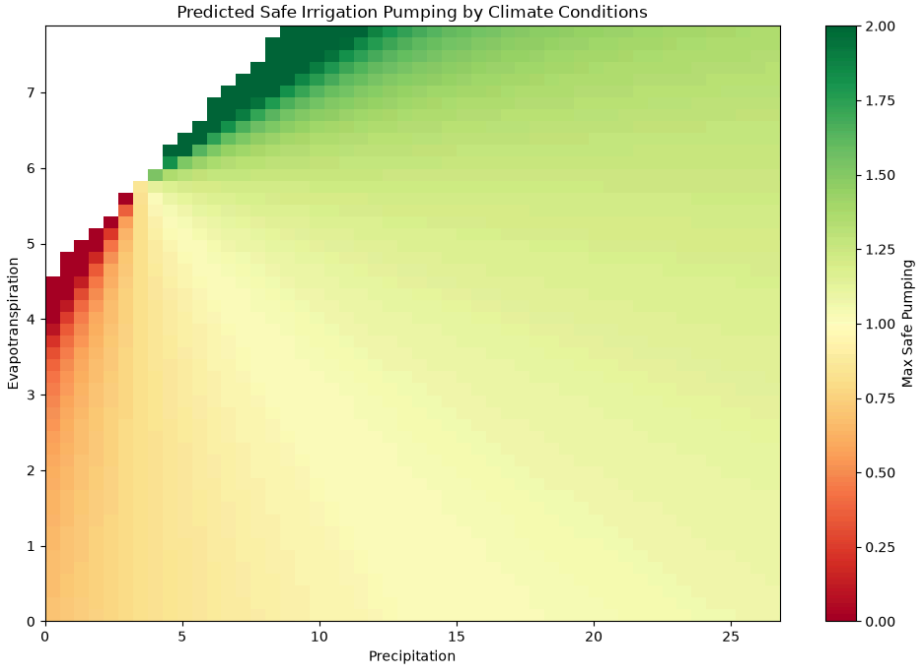
The regression model explains about 81% of the variation in baseflow ($R^2 = 0.8065$), which is strong for a hydrologic model. The most important finding is that irrigation pumping has a large negative effect on baseflow (coefficient = −5.11), meaning that for each unit increase in pumping, baseflow drops by about 5.11 units when other factors are held constant. Precipitation has a positive effect (+0.53), while evapotranspiration has a negative effect (−0.90). The interaction terms reveal that pumping's damage is worse during wet years (Pumping × Precipitation = −0.42) but less severe when ET is high (Pumping × ET = +1.11), likely because high-ET periods correspond to growing seasons when recharge patterns differ.



We validated the model using an 80/20 train-test split, and the test R^2 was 0.8045 with an RMSE of 25.36, confirming the model generalizes well. Using the regression equation, we derived safe pumping thresholds that vary by climate conditions. For example, under average conditions (Precipitation = 0.5, ET = 0.5), the safe pumping threshold is about 0.157 units, but it rises to 0.184 in dry/low-ET conditions and drops to 0.131 in wet/high-ET conditions. These dynamic thresholds offer water managers a practical tool for setting pumping limits that respond

to actual weather rather than fixed caps.





Technical

The RRCA baseflow CSV was loaded with date parsing. No missing values were present, so no imputation was needed. We engineered two interaction features (Pumping × Precipitation and Pumping × ET) to capture how climate moderates pumping's effect. All features were used in

their original scale without additional normalization, since OLS handles mixed-scale predictors and the coefficients remain interpretable.

Multiple linear regression was chosen because the response variable (baseflow) is continuous, the predictors are continuous and have plausible linear relationships with the response, and the goal is both prediction and interpretation. OLS via statsmodels provided detailed coefficient statistics (p-values, confidence intervals, F-statistic), while scikit-learn's LinearRegression was used for train/test validation.

We started with exploratory data analysis such as scatter plots, correlation matrices, time series, and spatial maps to understand the data structure before modeling. Our first regression used only the three main predictors without segment dummies, which gave a lower R^2. Adding segment fixed effects and interaction terms improved the model substantially. We validated with an 80/20 train-test split and checked residual diagnostics (residuals vs fitted, distribution of residuals, actual vs predicted) to confirm the model assumptions were reasonable.